

MODEL REFERENCE SHEET

Artificial Analysis benchmark snapshot

18 JUL 2026

13 supplied charts

GENERAL INTELLIGENCE Claude Fable 5 60	AGENTIC GPT-5.6 Sol max 54.0	CODING GPT-5.6 Sol xhigh 78.3
OUTPUT SPEED GPT-5.6 Luna max 185 tok/s	FIRST TOKEN Luna NR / Terra NR 0.7 s	500-TOKEN E2E GPT-5.6 Luna NR 3.5 s
LOWEST COST GPT-5.6 Luna low \$0.04/task	EARLY WRITING Claude Fable 5 max ~2,810 Elo	2D FRONTIER Luna -> Sol -> Fable cost vs. intelligence

Quick routing guide		
Use case	Primary pick	Why
Maximum general quality / writing	Claude Fable 5	Top Intelligence Index (60) and early writing result (~2,810 Elo); expensive (~\$2.70/task) and slow (124.5 s E2E).
Maximum agentic capability	GPT-5.6 Sol max	Agentic leader (54.0); also 59 intelligence and 77.4 coding. Use only when 138.6 s E2E is acceptable.
Highest coding score	GPT-5.6 Sol xhigh	Coding leader (78.3) at \$0.68/task and 51.1 s E2E.
Best coding value	GPT-5.6 Sol medium	76.3 coding at \$0.31/task and 14.3 s E2E; max adds only 1.1 coding points for 3.35x cost.
High-throughput reasoning	GPT-5.6 Luna max	185 tok/s and 51 intelligence; decode is fast, but reasoning makes TTFT 79.5 s.
Interactive balanced default	GPT-5.6 Terra high	49 intelligence, 119 tok/s, \$0.24/task, 1.8 s TTFT, 6.1 s E2E.
Lowest-cost strong reasoning	GPT-5.6 Luna xhigh	49 intelligence at \$0.14/task and 174 tok/s; accept 29.7 s TTFT.
Fast simple-task mode	GPT-5.6 Luna non-reasoning	0.7 s TTFT, 3.5 s E2E, 178 tok/s, \$0.05/task; intelligence falls to 27.
Balanced cross-vendor option	Grok 4.5 high	54 intelligence at \$0.31/task, 88 tok/s and 16.3 s E2E; stronger agentic than Sol medium, weaker coding.

NR = non-reasoning. "Best value" and routing labels are derived judgments from the supplied metrics, not source-chart categories. Values marked ~ elsewhere are estimates from cropped or unlabeled chart elements.

Full visible index rankings

Compact names: Sol/Terra/Luna imply GPT-5.6; "5.5" implies GPT-5.5; NR = non-reasoning. Ties retain the left-to-right order shown in the screenshots.

#	Agentic model	Score	#	Coding model	Score	#	Intelligence model	Score
1	Sol max	54.0	1	Sol xhigh	78.3	1	Fable 5 fallback	60
2	Fable 5 fallback	52.8	2	Sol max	77.4	2	Sol max	59
3	Sol xhigh	51.8	3	Sol high	77.2	3	Sol xhigh	58
4	Sol high	48.5	4	Terra max	76.7	4	Sol high	56
5	Terra max	47.4	5	Fable 5 fallback	76.5	5	Opus 4.8 max	56
6	Opus 4.8 max	47.2	6	Sol medium	76.3	6	Terra max	55
7	Sonnet 5 max	46.7	7	5.5 xhigh	74.9	7	5.5 xhigh	55
8	Grok 4.5 high	45.7	8	Opus 4.8 max	74.3	8	Grok 4.5 high	54
9	Luna max	45.6	9	Grok 4.5 high	72.4	9	Sol medium	54
10	5.5 xhigh	44.9	10	5.5 high	71.6	10	Sonnet 5 max	53
11	Terra xhigh	44.7	11	Sonnet 5 max	71.5	11	5.5 high	53
12	Sol medium	44.5	12	5.5 medium	71.5	12	Terra xhigh	52
13	5.5 high	43.5	13	Luna max	71.4	13	Luna max	51
14	Luna xhigh	42.9	14	Terra xhigh	70.6	14	5.5 medium	50
15	Terra high	41.3	15	Sol low	69.7	15	Sol low	49
16	Luna high	40.1	16	Luna xhigh	68.6	16	Luna xhigh	49
17	Sol low	40.0	17	Terra high	67.1	17	Terra high	49
18	5.5 medium	37.8	18	Sonnet 5 NR	66.4	18	Luna high	46
19	Terra medium	37.0	19	Sol NR	65.1	19	Terra medium	46
20	Sol NR	34.9	20	Terra medium	64.7	20	5.5 low	43
21	Sonnet 5 NR	33.7	21	Luna high	63.3	21	Sonnet 5 NR	42
22	Luna medium	31.0	22	5.5 low	60.9	22	Sol NR	41
23	Terra low	30.6	23	Terra low	58.1	23	Terra low	40
24	5.5 low	30.4	24	5.5 NR	56.5	24	Luna medium	38
25	Terra NR	29.3	25	Terra NR	52.3	25	5.5 NR	35
26	5.5 NR	25.8	26	Luna medium	50.7	26	Terra NR	34
27	Luna NR	~22.0	27	Luna NR	~39.0	27	Luna NR	27

The screenshots show 27 ranked entries in each index. The rightmost Luna non-reasoning Agentic and Coding labels were cropped; they are recorded as approximately 22 and 39.

Model matrix - GPT-5.6 Luna and Terra

Luna maximizes decode throughput; Terra generally provides the stronger interactive quality/latency balance.

Model / mode	Intel	Agent	Code	Cost/task	tok/s	TTFT s	E2E 500 s	Reference note
GPT-5.6 Luna (Non-reasoning)	27	~22.0	~39.0	\$0.05	178	0.7	3.5	Fastest E2E; lowest quality tier.
GPT-5.6 Luna (low)	33	—	—	\$0.04	—	—	—	Cheapest 2D frontier point; perf not shown.
GPT-5.6 Luna (medium)	38	31.0	50.7	\$0.05	177	1.7	4.5	Very low cost; 177 tok/s; 4.5 s E2E.
GPT-5.6 Luna (high)	46	40.1	63.3	\$0.09	176	6.6	9.5	Good low-cost quality step-up.
GPT-5.6 Luna (xhigh)	49	42.9	68.6	\$0.14	174	29.7	32.6	49 intel at \$0.14; long thinking delay.
GPT-5.6 Luna (max)	51	45.6	71.4	\$0.21	185	79.5	82.2	Fastest decode; speed-quality quadrant.
GPT-5.6 Terra (Non-reasoning)	34	29.3	52.3	—	123	0.7	4.8	0.7 s TTFT; cost not visible.
GPT-5.6 Terra (low)	40	30.6	58.1	\$0.10	115	1.3	5.7	Cheap, interactive baseline.
GPT-5.6 Terra (medium)	46	37.0	64.7	\$0.13	120	1.3	5.5	46 intel with 5.5 s E2E.
GPT-5.6 Terra (high)	49	41.3	67.1	\$0.24	119	1.8	6.1	Interactive sweet spot at 49 intel.
GPT-5.6 Terra (xhigh)	52	44.7	70.6	\$0.33	121	16.9	21.1	52 intel; speed-quality quadrant.
GPT-5.6 Terra (max)	55	47.4	76.7	\$0.55	136	133.9	137.6	Strong indexes; 134 s thinking delay.

~ = visually estimated from a clipped chart label. — = the metric was not visible in the supplied screenshots. Cost/task is the Artificial Analysis weighted Intelligence Index task cost, not a token-list price.

Within-family step-up economics			
Upgrade	Quality gain	Cost / time multiplier	Interpretation
Luna xhigh -> max	+2 intel; +2.7 agent; +2.8 code	1.50x cost; 2.52x E2E	Max mainly buys a small quality lift; xhigh is the stronger cost point.
Terra xhigh -> max	+3 intel; +2.7 agent; +6.1 code	1.67x cost; 6.52x E2E	Max meaningfully helps coding, but thinking time explodes from 16.9 s to 133.9 s.
Terra NR -> medium	+12 intel; +7.7 agent; +12.4 code	E2E only 4.8 -> 5.5 s	Medium is a compelling interactive reasoning step-up when cost permits.

Model matrix - GPT-5.6 Sol and GPT-5.5

Sol dominates the top agentic/coding ranks. Medium is the practical coding default; max is an agentic-quality mode with extreme thinking latency.

Model / mode	Intel	Agent	Code	Cost/task	tok/s	TTFT s	E2E 500 s	Reference note
GPT-5.6 Sol (Non-reasoning)	41	34.9	65.1	\$0.20	56	1.0	9.9	Simple tasks; same cost as Sol low.
GPT-5.6 Sol (low)	49	40.0	69.7	\$0.20	51	3.3	13.0	+8 intel vs NR for same \$0.20 cost.
GPT-5.6 Sol (medium)	54	44.5	76.3	\$0.31	53	4.7	14.3	Coding/value sweet spot.
GPT-5.6 Sol (high)	56	48.5	77.2	\$0.45	~49	12.1	22.3	High quality; moderate E2E.
GPT-5.6 Sol (xhigh)	58	51.8	78.3	\$0.68	53	41.6	51.1	Highest Coding Index.
GPT-5.6 Sol (max)	59	54.0	77.4	\$1.04	54	129.3	138.6	Highest Agentic Index.
GPT-5.5 (Non-reasoning)	35	25.8	56.5	\$0.17	66	0.9	8.4	Low latency; low index scores.
GPT-5.5 (low)	43	30.4	60.9	\$0.21	66	1.8	9.4	Small cost/latency step from NR.
GPT-5.5 (medium)	50	37.8	71.5	\$0.41	68	7.4	14.8	50 intel; eclipsed by Sol medium on value.
GPT-5.5 (high)	53	43.5	71.6	\$0.67	70	22.3	29.4	53 intel; 29.4 s E2E.
GPT-5.5 (xhigh)	55	44.9	74.9	\$0.99	68	61.6	69.0	55 intel; \$0.99; 69 s E2E.

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			Sol scale-up: coding saturates before agentic capability		
Mode	Intel	Agentic	Coding	Cost / E2E	Decision cue
low	49	40.0	69.7	\$0.20 / 13.0 s	Same \$0.20 cost as NR, but +8 intelligence.
medium	54	44.5	76.3	\$0.31 / 14.3 s	76.3 coding at \$0.31 and 14.3 s: best coding value.
high	56	48.5	77.2	\$0.45 / 22.3 s	Adds general quality with tolerable 22.3 s E2E.
xhigh	58	51.8	78.3	\$0.68 / 51.1 s	Absolute coding leader; 51.1 s E2E.
max	59	54.0	77.4	\$1.04 / 138.6 s	Agentic leader; only +1.1 coding vs medium at 3.35x cost.

Model matrix - Anthropic and xAI

The Sonnet effort variants are fully visible for speed/latency, but only max and non-reasoning are visible in the index/cost screenshots.

Model / mode	Intel	Agent	Code	Cost/task	tok/s	TTFT s	E2E 500 s	Reference note
Claude Sonnet 5 (Non-reasoning)	42	33.7	66.4	\$0.37	63	1.3	9.2	Fast; much lower scores than max.
Claude Sonnet 5 (low)	—	—	—	—	60	1.6	9.8	Latency/speed only in supplied views.
Claude Sonnet 5 (medium)	—	—	—	—	64	2.5	10.3	Latency/speed only in supplied views.
Claude Sonnet 5 (high)	—	—	—	—	65	8.5	16.2	Latency/speed only in supplied views.
Claude Sonnet 5 (xhigh)	—	—	—	—	68	22.7	30.0	Latency/speed only in supplied views.
Claude Sonnet 5 (max)	53	46.7	71.5	\$1.53	83	~193.7	~199.7	53 intel; about 200 s E2E.
Claude Opus 4.8 (max)	56	47.2	74.3	\$1.80	57	34.5	43.2	Premium quality; \$1.80/task.
Claude Fable 5 (with fallback)	60	52.8	76.5	~\$2.70	61	116.2	124.5	Top intelligence/writing; highest cost.
Grok 4.5 (high)	54	45.7	72.4	\$0.31	88	10.6	16.3	54 intel at \$0.31; 88 tok/s.

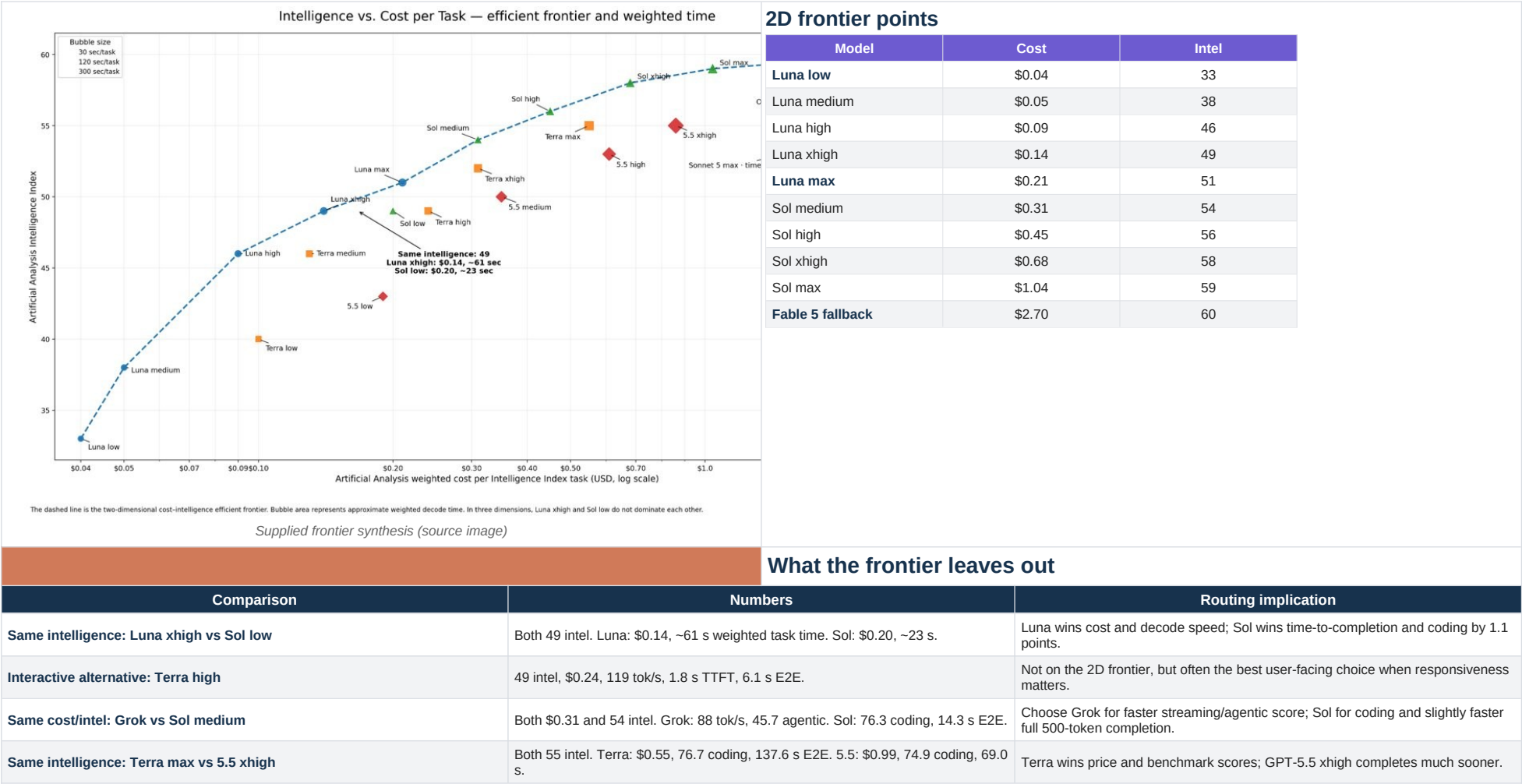
~ = visually estimated from a clipped chart label. — = the metric was not visible in the supplied screenshots. Cost/task is the Artificial Analysis weighted Intelligence Index task cost, not a token-list price.

				Performance and cost leaderboards			
#	Output speed	tok/s	#	E2E 500 tokens	seconds	#	Lowest cost / task
1	Luna max	185	1	Luna NR	3.5	1	Luna low \$0.04
2	Luna NR	178	2	Luna medium	4.5	2	Luna NR \$0.05
3	Luna medium	177	3	Terra NR	4.8	3	Luna medium \$0.05
4	Luna high	176	4	Terra medium	5.5	4	Luna high \$0.09
5	Luna xhigh	174	5	Terra low	5.7	5	Terra low \$0.10
6	Terra max	136	6	Terra high	6.1	6	Terra medium \$0.13

Output speed measures decode throughput after answering begins. It does not remove reasoning delay: Luna max leads at 185 tok/s yet takes 79.5 s to the first answer token.

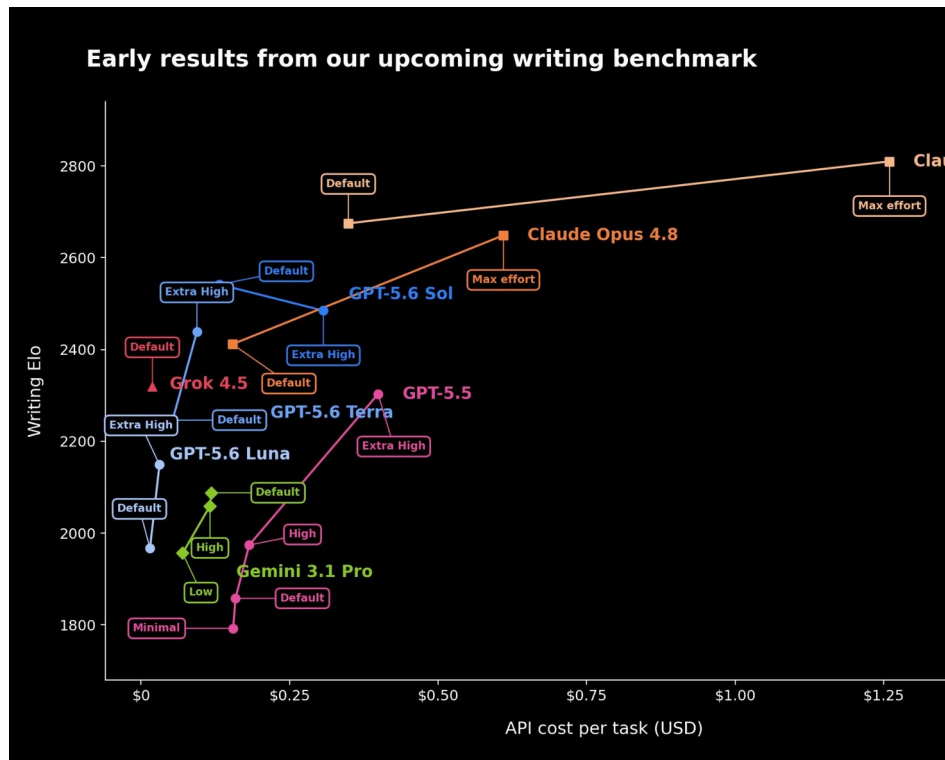
Cost-intelligence frontier and the third dimension: time

The supplied synthesis chart treats bubble area as approximate weighted task time. The dashed line is the 2D cost-intelligence efficient frontier.



Early writing benchmark: approximate ranking

The source labels this benchmark as “early results.” Elo and API cost points below are visually read from the axes and rounded; they are not final benchmark values.



Approximate top results

#	Setting	Cost	Elo
1	Claude Fable 5 - max effort	~\$1.26	~2,810
2	Claude Fable 5 - default	~\$0.35	~2,675
3	Claude Opus 4.8 - max effort	~\$0.61	~2,650
4	GPT-5.6 Sol - default	~\$0.17	~2,550
5	GPT-5.6 Sol - extra high	~\$0.31	~2,485
6	GPT-5.6 Terra - extra high	~\$0.10	~2,440
7	Claude Opus 4.8 - default	~\$0.16	~2,410
8	Grok 4.5 - default	~\$0.02	~2,315
9	GPT-5.5 - extra high	~\$0.40	~2,300
10	GPT-5.6 Luna - extra high	~\$0.03	~2,150

Writing-specific takeaways

- Claude Fable 5 leads: max effort is about 2,810 Elo; default is about 2,675.
- Fable max costs about 3.6x default for roughly +135 Elo.
- Claude Opus 4.8 max reaches about 2,650 Elo at about \$0.61/task.
- Grok 4.5 default is unusually inexpensive (~\$0.02) for roughly 2,315 Elo.

- GPT-5.6 Sol default appears to beat its extra-high setting while costing less; treat this non-monotonic result as provisional.
- GPT-5.6 Terra extra high reaches roughly 2,440 Elo at about \$0.10/task.
- Writing effort labels do not necessarily map one-to-one to the index-chart effort labels.
- Writing API cost/task is a different workload from Intelligence Index cost/task; do not compare the dollar values directly.

Metric definitions, cost anatomy, and transcription notes

Use this page when applying the sheet to routing, procurement, or benchmark interpretation.

Definitions Intelligence Index v4.1: Composite of 9 evaluations: GDPval-AA v2, tau^3-Banking, Terminal-Bench v2.1, SciCode, Humanity's Last Exam, GPQA Diamond, CritPt, AA-Omniscience, and AA-LCR. Agentic Index: Average of agentic-capability benchmarks in the source: GDPval-AA v2 and tau^3-Banking. Coding Index: Weighted average of Terminal-Bench v2.1 and SciCode. Output speed: Output tokens per second after generation begins; higher is better. TTFT: Seconds to first answer token, including reasoning-model thinking time; lower is better. E2E 500: Seconds to produce 500 output tokens, including thinking and input processing; lower is better. Cost/task: Weighted average USD per Artificial Analysis Intelligence Index task; lower is better. Writing Elo: Separate early writing benchmark rating plotted against API cost per writing task.		Cost anatomy in the supplied charts - Token-type stack: answer, reasoning, cache write, cache hit, and input. - Evaluation stack: GDPval-AA v2, tau^3-Banking, Terminal-Bench v2.1, SciCode, Humanity's Last Exam, GPQA Diamond, CritPt, AA-Omniscience, and AA-LCR. - At the high-cost end, GDPval-AA v2 is visibly a major contributor; Terminal-Bench is also material for Claude Opus/Fable. - Use the total bar labels as the primary comparable number. Several segment labels are not visible in the screenshot. Interpretation cautions - Index values are aggregate scores, not percentages or guaranteed task success rates. - The UI screenshots are horizontally scrollable; some selected models may be outside the captured viewport. - Latency is environment- and provider-dependent. Treat it as the measured snapshot, not a service-level guarantee. - Fable is "with fallback" in index charts, while the writing chart uses default/max effort labels; these may not be identical routes. - Reasoning modes are indicated by a lightbulb in the source charts. Non-reasoning rows are explicitly labeled NR here. - The weighted task-time bubbles in the frontier chart are not the same measure as the 500-token E2E bars.	
		Source image coverage	
Supplied image		What was extracted	
Artificial Analysis Agentic Index		Full 27-entry agentic ranking and scores.	
Artificial Analysis Coding Index		Full 27-entry coding ranking and scores.	
Artificial Analysis Intelligence Index		Full 27-entry intelligence ranking and benchmark composition.	
Intelligence vs. Output Speed		Speed-quality quadrant and model trade-offs.	
End-to-End Response Time		500-token completion times and thinking/output decomposition.	
Latency - Time To First Answer Token		First-token latency including reasoning time.	
Output Speed		Decode throughput in output tokens per second.	
Evaluation Breakdown		Cost contribution by each Intelligence Index evaluation.	
Cost per Intelligence Index Task		Total cost and token-type cost stack.	
Intelligence vs. Cost per Intelligence Index Task		Cross-vendor cost-quality quadrant.	
HM8OUrWWMAAkIO1.jpeg		Early writing Elo versus API task cost.	
HM6BgeuaMAA6XR8.jpeg		OpenAI family cost-intelligence curves and attractive quadrant.	
bench.jpeg		2D efficient frontier plus weighted task-time bubbles.	

Transcription rule: exact printed labels were copied where visible; values prefixed with ~ were estimated from chart geometry or a cropped label. No external/live data was mixed into this reference sheet.